ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

Version 1.0 · January 2026*

Non-Confidential

© 2026 TOBE ENERGY INC.

EXECUTIVE SUMMARY

Tobe Energy has developed a fundamentally different approach to water electrolysis that achieves **>92% system efficiency (HHV)** while eliminating the exotic materials, complex thermal management, and degradation mechanisms that constrain conventional electrolyzer technologies.

Key Performance Metrics:

- **94.7% Stack Electrical Efficiency (LHV)**
- **>92% System Efficiency (HHV)**
- **~75% CAPEX reduction** vs. PEM at scale
- **1,000+ hours** tested
- **80,000+ est. stack life (hrs)**

Key Differentiators

ATTRIBUTE	CONVENTIONAL ELECTROLYZERS	TOBE APPROACH
Operating Temperature	60–80°C (PEM), 70–90°C (Alkaline)	Near ambient (<30°C)
Critical Materials	Platinum, Iridium, Nafion membranes	304 SS, commodity components
Electrolyte	Concentrated KOH (25–40%)	Dilute solution (pure water compatible)
Primary Failure Mode	Membrane degradation, catalyst poisoning	Standard industrial wear items
Thermal Management	Active cooling required	Minimal (isothermal operation)

The Core Insight

Conventional electrolysis generates significant waste heat because it operates continuously at voltages well above thermodynamic minimum. Tobe's pulsed approach works at lower effective overpotentials, converting electrical energy to chemical energy with minimal thermal loss.

THE PROBLEM

Green hydrogen remains expensive primarily because electrolysis is inefficient. Electricity accounts for **60–80% of levelized hydrogen cost**, yet conventional electrolyzers convert only **60–75% of input energy** to hydrogen.

Why Conventional Systems Are Stuck

PEM Electrolyzers

© 2026 TOBE ENERGY INC.

- Achieve good current density but require platinum-group metals and expensive proton-exchange membranes
- Membrane degradation limits lifetime
- Iridium scarcity constrains scale-up — global production cannot support terawatt-scale deployment

Alkaline Electrolyzers

- Avoid PGMs but require concentrated caustic electrolyte (25–40% KOH), creating safety, handling, and corrosion challenges
- Limited turndown capability makes them poorly suited for variable renewable power

The Thermal Penalty

Both technologies operate at elevated temperatures (60–90°C) and generate substantial waste heat. This isn't incidental — it's fundamental to how they work.

CONVENTIONAL APPROACH	TOBE APPROACH
Continuous DC operation at 1.8–2.2V	Pulsed waveform operation
High overpotentials → waste heat	Lower effective overpotentials
Active cooling infrastructure required	Near-isothermal operation
65–75% system efficiency typical	**>92% system efficiency demonstrated**
Balance of plant adds cost and failure modes	Simplified system architecture

The Scale Challenge

Meeting global decarbonization targets requires electrolyzer capacity measured in terawatts. Current PGM-dependent technologies face fundamental material constraints. A new approach using commodity materials isn't just cost-competitive — it's necessary for scale.

WHAT TOBE IS

Tobe is a category-level rethink of water electrolysis. Rather than incremental improvements to PEM or alkaline systems, we've developed a fundamentally different electrochemical architecture that eliminates the constraints those technologies accept as given.

Core Architecture

© 2026 TOBE ENERGY INC.

SUBSYSTEM	FUNCTION	KEY INNOVATION
Power Electronics	Converts input to controlled pulse waveforms	Proprietary pulse patterns for electrochemical kinetics
Electrolysis Module	Houses electrode stacks and manages electrolyte	Membrane-free design using 304 SS construction
Gas Handling	Separates, dries, and pressurizes H_2 and O_2	Simplified separation (no membrane → cleaner streams)

What Makes It Different

- **No Membranes** — Zero proton-exchange membrane
- **No Caustic** — Works with dilute or pure water
- **No PGMs** — Zero platinum or iridium
- **Near Ambient** — Operates <30°C, no active cooling

Where It Fits

- Behind-the-meter installations — Industrial facilities with variable renewable generation
- Hydrogen hubs — Centralized production serving multiple offtakers
- Data center backup power — Converting excess renewable capacity to stored hydrogen
- Distributed production — Small-scale on-site generation for fleet fueling or industrial processes

---

PERFORMANCE

We report efficiency using clear definitions to enable direct comparison. All figures represent measured system performance, not theoretical or stack-only values.

LHV vs. HHV Basis

Hydrogen's lower heating value (LHV) is **33.33 kWh/kg**. The higher heating value (HHV) is **39.41 kWh/kg**. We report LHV efficiency as the industry standard for combustion and fuel cell applications.

Measured Performance

PARAMETER	VALUE	NOTES
Stack Electrical Efficiency (LHV)	**94.7%**	Stack performance at nominal operating point
System Efficiency	**~42.23 kWh/kg H_2**	Including power electronics and gas handling
Turndown Range	**10–100%**	Maintains efficiency across operating range
Response Time	**<1 second**	Grid-following capability for renewable integration
Startup Time	**Immediate**	No warm-up period required
Operating Pressure	**Up to 350 psig**	Reduces downstream compression
Operating Temperature	**<30°C**	Near-ambient, minimal thermal management
Ambient Range	**-30°C to +50°C**	Outdoor installation without climate control
H_2 Purity	**>99%**	Higher purity available with standard PSA

Why This Matters

At typical industrial electricity rates ($0.04–0.06/kWh), improving efficiency from 70% to 94% reduces the electricity cost component of hydrogen from ~$2.38/kg to ~$1.77/kg. Over the lifetime of a facility producing **1,000 kg/day**, this represents millions in savings.

RELIABILITY

Tobe's architecture eliminates the primary failure modes that limit conventional electrolyzer lifetime. Nothing in the stack gets "used up" during operation.

Failure Modes Eliminated

CONVENTIONAL FAILURE MODE	ROOT CAUSE	TOBE APPROACH
Membrane Degradation	Chemical attack, mechanical stress	No membrane in system
Catalyst Poisoning	Contamination of PGM catalysts	No catalysts required
Thermal Cycling Fatigue	Expansion/contraction from temp swings	Isothermal operation
Caustic Corrosion	Concentrated KOH attacks materials	Dilute/neutral electrolyte
Electrode Degradation	Corrosion at high temperature	Low-temp with corrosion-resistant 304 SS

© 2026 TOBE ENERGY INC.

Lifetime Projection

- **80,000+ estimated stack hours**
- Wear items only — standard replacement schedule
- Serviceable design for field maintenance
- Minimal scheduled downtime

Design Philosophy

- **Commodity materials** — 304 stainless steel uses globally available supply chains
- **Modular architecture** — Individual cells serviceable without system shutdown
- **No single points of failure** — Redundant sensing and control throughout
- **Industrial standards** — Components selected for proven reliability

---

SCALING APPROACH

Tobe systems scale through modular multiplication of proven cell designs. This maintains performance while reducing manufacturing risk and enabling flexible deployment.

Product Line

Note: The model designations below reflect technology spec sheet nomenclature. Commercial product names are T-25 (50 kW) and T-125 (250 kW).

MODEL	CAPACITY	H_2 OUTPUT	FOOTPRINT	TARGET APPLICATION
Tobe-25	25 kW	~12 kg/day	Compact skid	Distributed, pilot, R&D
Tobe-250	250 kW	~120 kg/day	20' container	Commercial, fleet fueling
Tobe-2500	2.5 MW	~1,200 kg/day	Multi-container	Industrial, hub-scale

Manufacturing Philosophy

© 2026 TOBE ENERGY INC.

Current State

- Prototype systems built using standard fabrication
- No specialized equipment or exotic materials required
- Current cost includes NRE and low-volume inefficiencies
- Prototype cost: **~$446/kW**

Production Projection

- At MW-scale volumes, cost converges toward commodity manufacturing
- Primary drivers are power electronics and structural steel — well-understood supply chains
- Projected at scale: **~$177/kW**

CAPEX Advantage

Conventional PEM electrolyzers cost **$700–1,200/kW** at scale. Alkaline systems range **$500–800/kW**. Tobe's projected **$177/kW** represents ~75% reduction from PEM and ~65% from alkaline.

Supply Chain Considerations

- **No PGM dependency** — Eliminates platinum/iridium price volatility and supply constraints
- **No membrane supply chain** — Removes Nafion sourcing as bottleneck
- **Standard industrial components** — Leverages global 304 SS and power electronics base

---

VALIDATION STATUS

Tobe has completed extensive prototype testing under DOE SBIR grant funding, with independent measurement of key performance parameters across multiple operating conditions.

Testing Summary

TEST CAMPAIGN	SYSTEM	DURATION	CONDITIONS	KEY RESULTS
Phase 1	6 kW	500+ hrs	Low TDS water	**94.2% efficiency confirmed**
Phase 2	6 kW	300+ hrs	High TDS water	Consistent, no degradation
Phase 3	15 kW	200+ hrs	Variable load	Full turndown validated

© 2026 TOBE ENERGY INC.

Measurement Methodology

- **Power measurement**: INA260 precision power sensors (±1% accuracy)
- **Flow measurement**: FS1015 mass flow transmitters with NIST-traceable calibration
- **Purity verification**: KE-25 electrochemical O_2 sensors for cross-gas detection
- **Safety monitoring**: TGS821 hydrogen leak detection throughout facility

What Remains

Validated

- Core efficiency claims (94.7% stack electrical, LHV)
- Turndown capability (10–100%)
- Water quality tolerance
- Response time and startup behavior
- Gas purity at output

In Progress

- Extended duration testing (8,760 hrs target)
- Full environmental range (-30°C to +50°C)
- Pressure cycling endurance
- Production unit cost validation
- Third-party certification pathway

NEXT STEPS

Tobe is advancing from prototype validation to pilot deployment, with a clear pathway to commercial production for qualified early adopters.

Development Roadmap

PHASE	TIMELINE	MILESTONE
Prototype Validation	Complete	Core performance demonstrated at 6–15 kW scale
Pilot Unit Build	Q1–Q2 2026	25 kW integrated system for field deployment
Pilot Deployment	Q3 2026	Installation at partner site with operational data
Commercial Unit	2027	250 kW containerized system for early commercial sales

© 2026 TOBE ENERGY INC.

Engagement Opportunities

- **Pilot Partners** — Host sites for field validation
- **Strategic Partners** — Integration into larger systems
- **Investors** — Seed round for commercial scale-up
- **Technical Review** — Detailed diligence for qualified parties

Contact

Ready to explore how Tobe can support your hydrogen strategy? Email: info@tobe.energy Web: tobe.energy Request: Technical specs, pilot scoping call, or diligence materials

Disclaimer: This document contains forward-looking statements regarding projected performance, costs, and timelines. Actual results may vary. Performance figures represent laboratory measurements under controlled conditions; field performance may differ. Cost projections assume production volumes not yet achieved.

© 2026 Tobe Energy Inc. | tobe.energy